

(A) Clinicians Input

“Design a trial for Head and Neck
Squamous Cell Carcinoma with xxxxx”

or

Eligibility Criterion draft for HNSCC
Inclusion criteria Exclusion criteria



(B) Trials Retrival

Retrieve all relavent trials
using API

ClinicalTrials.gov

Start Date
Completion Date
Sites Count
Enrollment Count
Eligibility
.....

(C) General EC Drafting

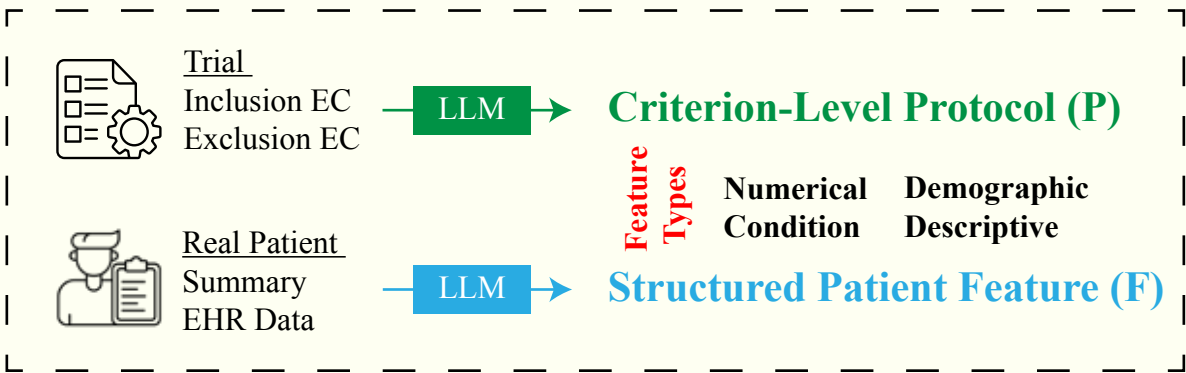
Past enrollment statistics
Recommendation from existing trials
Mortality/Adverse event/Enrollment Prediction

(D) Criterion-level fine-tuning

Criterion recommendation and analysis
Cross-trial statistics by rule similarity
Leave-one-out analysis on Part C prediction



Features Extraction

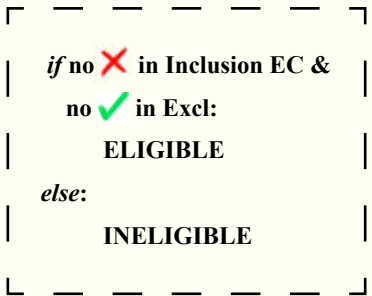


SBERT Semantic Encoder

Features Alignment

	P1	P2	P3	P4	P5	P6
F1	0.96	0.45	0.09	0.35	0.01	0.07
F2	0.62	0.01	0.12	0.67	0.79	0.01
F3	0.12	0.22	0.55	0.11	0.72	0.03
F4	0.50	0.12	0.36	0.74	0.72	0.06
F5	0.24	0.51	0.49	0.09	0.22	0.01
F6	0.02	0.00	0.11	0.31	0.19	0.09

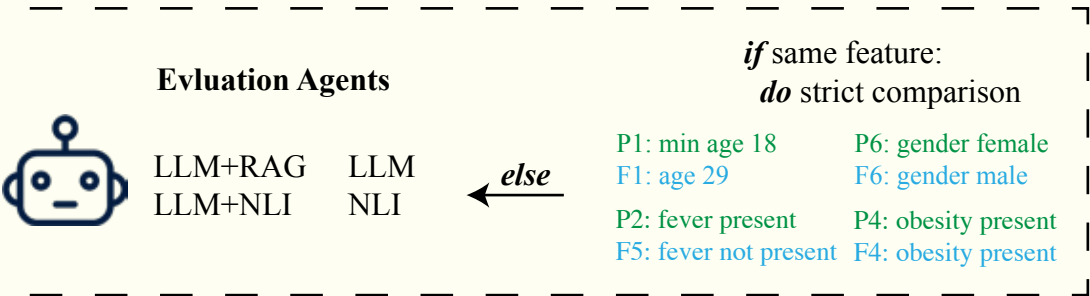
Rule Based Decision



Interpretable Results

P1: F1 ✗
P2: F5 ✓
P3: F3 ✗
P4: F4 ✗
P5: F2 ✓
P6: F6 ✓

Criterion Level Evaluation



Patient Trial Matching

Quantification of Trial/Rule Exclusion Rate

Protocols Interaction

DATA-DRIVEN AI TRIAL DESIGNER WITH REAL-WORLD SIMULATION